

gravity - practice problems

name _____

1. An asteroid in the main belt has an average orbital distance of 2.25 AU from the Sun. Using Kepler's simple relation, find its orbital period in years.
2. A comet takes 8 years to orbit the Sun. Using Kepler's simple relation, find its average orbital distance in AU.
3. Two 100 000 kg pieces of mining equipment are parked 20 m apart, measured from center to center. What is the gravitational force between them?
4. A 1200 kg satellite is 400 km above Earth's surface, about the altitude of some low Earth orbit spacecraft. Find the force of gravity on the satellite. Use Earth's mass as 5.97×10^{24} kg and Earth's radius as 6.37×10^6 m.
5. A 75 kg astronaut feels a 300 N gravitational force from Earth. How far is the astronaut from Earth's center? Use Earth's mass as 5.97×10^{24} kg.

6. A 75 kg person stands on Earth. Earth pulls on the person, and the person pulls on Earth with the same size force. Find the person's acceleration and Earth's acceleration from this gravitational force. Use Earth's mass as 5.97×10^{24} kg and Earth's radius as 6.37×10^6 m.

7. Mars has a mass of 6.42×10^{23} kg and a radius of 3.39×10^6 m. Find the gravitational acceleration at the surface of Mars.

8. At what altitude above Earth's surface is gravitational acceleration one-fourth of its surface value? Use Earth's radius as 6.37×10^6 m.

9. Europa, one of Jupiter's moons, has a radius of 1.56×10^6 m and orbits 6.7×10^8 m from Jupiter's center. What is the gravitational acceleration on Europa's surface?

10. A 70 kg rover part is weighed on Mars. Use the surface gravity you found for Mars, 3.73 m/s^2 . What is its weight on Mars?

11. An astronaut on the Moon drops a hammer from 1.5 m above the ground. It lands 1.36 s later. The Moon's radius is 1.74×10^6 m. Use these measurements to find the Moon's mass.
12. A rock is dropped from rest near the surface of Mars. Assume the gravitational acceleration stays constant at 3.73 m/s^2 during the short fall. How fast is the rock moving after 6.0 s?
13. A 60 kg hiker climbs 1000 m up a mountain. Find the change in gravitational potential energy using $U = -GMm/r$, and compare it to mgh from the energy page. Use Earth's mass as 5.97×10^{24} kg and Earth's radius as 6.37×10^6 m.
14. A 50 kg rock starts at rest 1000 km above the Moon's surface and falls straight down. There's no air on the Moon. How fast is it moving when it hits the surface? Use the Moon's mass as 7.35×10^{22} kg and its radius as 1.74×10^6 m.
15. A 1000 kg payload is lifted from Earth's surface to a circular-orbit altitude of 400 km. Ignoring the rocket's wasted energy, how much work is needed just to increase the payload's gravitational potential energy? Use Earth's mass as 5.97×10^{24} kg and Earth's radius as 6.37×10^6 m.

16. Find the minimum escape speed from the surface of Mars. Use Mars's mass as 6.42×10^{23} kg and its radius as 3.39×10^6 m.

17. A small asteroid has mass 8.0×10^{15} kg and radius 500 m. Find its surface gravity and escape speed.

18. A probe starts from rest at a distance of 2 Earth radii from Earth's center and falls to Earth's surface. Ignoring atmosphere and Earth's rotation, how fast is it moving when it reaches the surface? Use Earth's mass as 5.97×10^{24} kg and Earth's radius as 6.37×10^6 m.

19. Halley's Comet has an orbital period of about 75 years. If Kepler's simple relation is used as an average model for its orbit around the Sun, what is its average distance from the Sun in AU?

20. An example on this page found that the Sun pulls on Earth with a force of 3.56×10^{22} N. Use Earth's mass, 5.97×10^{24} kg, to find Earth's acceleration toward the Sun.